

Customer Behaviour Analysis

• Project Overview:

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

• Dataset Summary:

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
 - Missing Data: 37 values in Review Rating column

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

• Data Exploration using Python:

We began with data preparation and cleaning in Python:

- Data Loading: We imported the data using **pandas**.
- Data Exploration: Used **df.info().describe()** for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discour Applie
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	390
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	N
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	222
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	Na
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	Na
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	Na
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	Na
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	Na
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	Na
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	Na

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	purchase_amount	discount_appli...	
	2	64	Yes	
	3	73	Yes	
	4	90	Yes	
	7	85	Yes	
	9	97	Yes	
	12	68	Yes	
	13	72	Yes	
	16	81	Yes	
	20	90	Yes	
	22	62	Yes	
	24	88	Yes	
	29	94	Yes	
	32	79	Yes	
	33	67	Yes	
	35	81	Yes	

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchas...	Average Review	
	Gloves	3.86	
	Sandals	3.84	
	Boots	3.82	
	Hat	3.8	
	Skirt	3.78	

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type	round(avg(purchase_amount...
	Express	60.48
	Standard	58.46

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status	total_revenue	average_spend	total_customer	
	Subscribed	62645	59.4919	1053	
	Non-subscribed	170436	59.8651	2847	

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchas...	purchase_rate	
	Hat	50.0000	
	Sneakers	49.6552	
	Coat	49.0683	
	Sweater	48.1707	
	Pants	47.3684	

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segmn...	number of custom...	
	Loyal	3116	
	Returning	701	
	New	83	

8. **Top 3 Products per Category** – Listed the most purchased products within each category

	item_rank	category	item_purchas...	total_orders	
	1	Accessories	Jewelry	171	
	2	Accessories	Belt	161	
	3	Accessories	Sunglasses	161	
	1	Clothing	Pants	171	
	2	Clothing	Blouse	171	
	3	Clothing	Shirt	169	
	1	Footwear	Sandals	160	
	2	Footwear	Shoes	150	
	3	Footwear	Sneakers	145	
	1	Outerwear	Jacket	163	
	2	Outerwear	Coat	161	

9. **Repeated Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

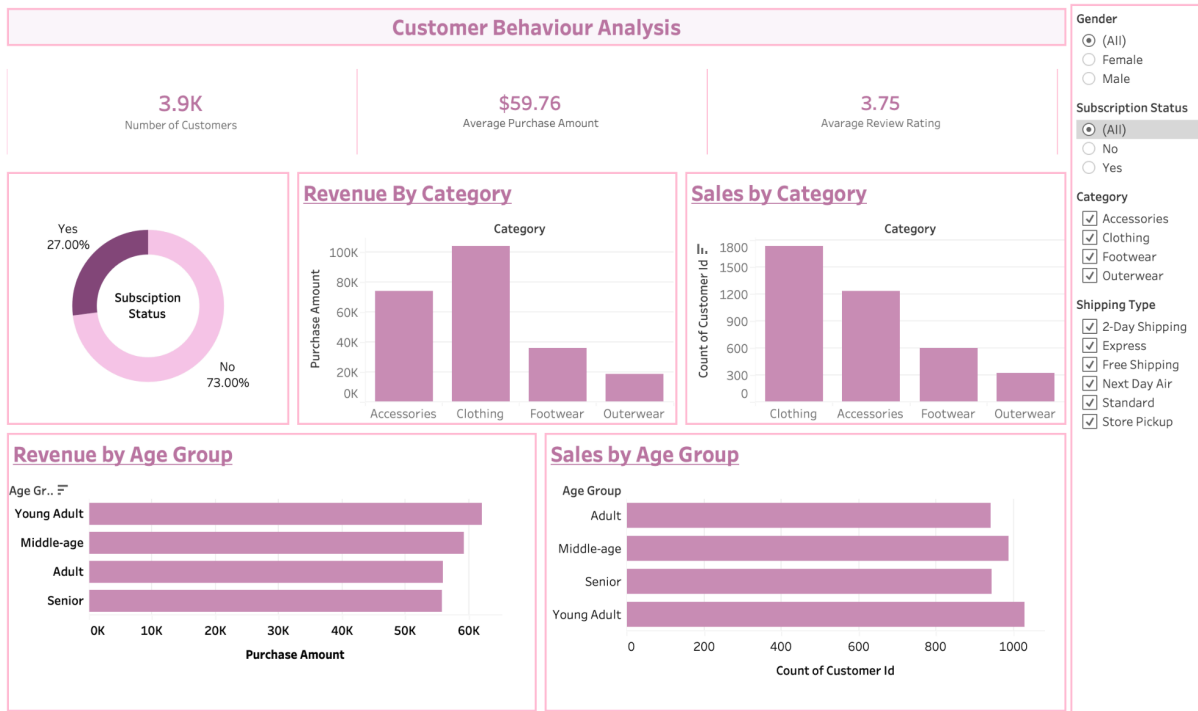
	subscription_sta...	repeated_buye...	
	Yes	958	
	No	2518	

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group	revenue	
	Young Adult	62143	
	Middle-age	59197	
	Adult	55978	
	Senior	55763	

Dashboard in Tableau:

Finally, we built an interactive dashboard in Tableau to present insights visually.



Business Recommendation:

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.